Ahmad Eyad Muhanna 2230006180

Lab2: Text Pre-processing and Regular Expressions

In this lab, you will learn essential techniques for preparing and cleaning text data using various pre-processing steps. Additionally, you will gain hands-on experience with regular expressions, a powerful tool for pattern matching and text manipulation in natural language processing (NLP) tasks.

Regular Expressions

Regular expressions (regex) are extremely useful in extracting information from any text by searching for one or more matches of a specific search pattern

https://www.regexpal.com/

Regular Expressions use cases

- Data pre-processing
- Pattern matching
- Text feature Engineering
- Web scraping
- Data validation
- Data extraction

An example use case is extracting all hashtags from a tweet, or getting email addresses or phone numbers from large unstructured text content.

Regular Expressions with Python

Python provides a convenient built-in module for managing regular expressions:

```
import re
```

We can import it as we import any other Python library.

Important symbols in Regular Expression

Let's start with the basic regular expression characters and some examples:

- (^) : Matches the expression to its right, at the start of a string before it finds a line break.
- ($) : Matches the expression to its left, at the end of a string before it finds a line break.
- (.) : Matches any character except newline.
- (a) : Matches exactly one character.
- (ab) : Matches the string ab.

Quantifiers:

- (a|b) : Matches expression a or b. If a is matched first, b is not checked;
- (+) : Matches the expression to its left 1 or more times;
- (*) : Matches the expression to its left 0 or more times;
- (?) : Matches the expression to its left 0 or 1 times.

Character Classes:

- \w : Matches alphanumeric characters, that is a-z, A-Z, 0–9, and underscore(_);
- \W : Matches non-alphanumeric characters, that is except a-z, A-Z, 0–9, and _;
- \d : Matches digits, from 0–9;
- \D : Matches any non-digits;
- \s : Matches whitespace characters, which also include the \t, \n, \r, and space characters;
- \S : Matches non-whitespace characters.
- \n : Matches a newline character;
- \t : Matches a tab character;
- \b : Matches the word boundary (or empty string) at the start and end of a word;
- \B : Matches where \b does not, that is non-word boundary.

Sets:

- [abc] : Matches either a, b, or c. It does not match abc;
- [a-z] : Matches any alphabet from a to z;
- [A-Z] : Matches any alphabets in capital from A to Z;
- [a-p] : Matches a, -, or p. It matches - because \ escapes it;
- [-z] : Matches - or z;
- [a-z0–9] : Matches characters from a to z or from 0 to 9.
- [(+*)] : Special characters become literal inside a set, so this matches (, +, *, or);
- [^ab5] : Adding ^ excludes any character in the set. Here, it matches characters that are not a, b, or 5;

Regular Expression functions

In [1]:
```
import re

text = "Woodchucks is another name for groundhog!"

re.findall(r"groundhog|woodchuck",text)
```

Out[1]:
```
['groundhog']
```

search

Search for pattern occurrences in a string using the search function of the **re module**. This function returns a match object, containing the first matched substring (or None, if it doesn't exist) and its position inside the original string.

In [2]:
```
import re

text = 'I am enjoying the NLP course.'

print(re.search(r"I", text)) #r"I" means it's a raw string literal
print(re.search(r"se.$", text)) #$ Search at the end of the string
print(re.search(r"am", text))
print(re.search(r"m", text))
print(re.search(r"AI", text))
```

```
<re.Match object; span=(0, 1), match='I'>
<re.Match object; span=(26, 29), match='se.'>
<re.Match object; span=(2, 4), match='am'>
<re.Match object; span=(3, 4), match='m'>
None
```

In [3]:
```
match = re.search((r"I"),text)
print(match.span())
print(match.start())
print(match.end())
print("The matched pattern is: ",text[match.start():match.end()]) #return matched s
print("The matched pattern is: ", match.group()) #return matched string using group
```

```
(0, 1)
0
1
The matched pattern is:  I
The matched pattern is:  I
```

match

The match function is similar to search, but it only tries to match the pattern at the beginning of the target string.

In [6]:
```
import re
re.match?
```

```
Signature: re.match(pattern, string, flags=0)
Docstring:
Try to apply the pattern at the start of the string, returning
a Match object, or None if no match was found.
File:      c:\users\user\appdata\local\programs\python\python313\lib\re\__init__.py
Type:      function
```

In [7]:
```
import re

text = 'I am enjoying the NLP course.'

print(re.search(r"I.*", text)) #global
print(re.match(r"I.*", text)) #local >> search at the start of the string

print(re.search(r"enjoying.*", text))
print(re.match(r"enjoying.*", text))
```

```
<re.Match object; span=(0, 29), match='I am enjoying the NLP course.'>
<re.Match object; span=(0, 29), match='I am enjoying the NLP course.'>
<re.Match object; span=(5, 29), match='enjoying the NLP course.'>
None
```

In [8]:
```
import re

text = '1999 was the year I was born.'

print(re.search(r"[a-zA-Z]+", text))
print(re.match(r"[a-zA-Z]+", text))
```

```
<re.Match object; span=(5, 8), match='was'>
None
```

Search does a global search, whereas match does a local search!

Match is often used when you need to check if the string starts with a specific pattern.

Search is used when you want to find a pattern anywhere within the string.

findall

The findall function looks for all the pattern matches in the target string (whereas search and match look only for the first occurrence).

- `re.findall()` returns a list of all matches.
- `re.finditer()` returns an iterator of match objects.
- `re.search()` and `re.match()` both return a single match object for the first occurrence (but differ in where they start matching: match() matches only at the start of the string, search() anywhere).

In [9]:
```
import re

text1 = "a 1 b 2 c 3"
```

```
text2 = "a1 b2 c 3"
text3 = "a 1 b 222 c 3"

print(re.findall(r"\d+", text1)) #match any digit
print(re.findall(r"\d+", text2))
print(re.findall(r"\d+", text3))
```

```
['1', '2', '3']
['1', '2', '3']
['1', '222', '3']
```

sub

The sub is a function that finds patterns in the target string and substitute them with another string.

flags=re.I --> non-case-sensitive

count = specify number of matches you want to replace

In [10]:
```
import re

re.sub?
```

```
Signature: re.sub(pattern, repl, string, count=0, flags=0)
Docstring:
Return the string obtained by replacing the leftmost
non-overlapping occurrences of the pattern in string by the
replacement repl.  repl can be either a string or a callable;
if a string, backslash escapes in it are processed.  If it is
a callable, it's passed the Match object and must return
a replacement string to be used.
File:      c:\users\user\appdata\local\programs\python\python313\lib\re\__init__.py
Type:      function
```

In [11]:
```
import re

text = 'I am enjoying the Math course.'

print(re.sub(r"Math", "NLP", text, count=1, flags=re.I)) #flags=re.I >> Ignore the
                                                         # count = 1 >> number of word I ne
```

```
I am enjoying the NLP course.
```

compile

The compile function compiles a regular expression into a regular expression object, which allows for caching and faster pattern matching.

In [12]:
```
import re
re.compile?
```

```
Signature: re.compile(pattern, flags=0)
Docstring: Compile a regular expression pattern, returning a Pattern object.
File:      c:\users\user\appdata\local\programs\python\python313\lib\re\__init__.py
Type:      function
```

In [13]:
```
import re

text = "This year is 2021"
pattern = re.compile(r"\d+")
print(type(pattern))

print(pattern.sub("2022", text))
```

```
<class 're.Pattern'>
This year is 2022
```

split

The split function is similar to the Python split function, but splits according to regular expression patterns.

In [14]:
```
string = 'Apple,Orange,Mango,Banana'
print(string.split(','))
```

```
['Apple', 'Orange', 'Mango', 'Banana']
```

In [15]:
```
import re

text = "a 1 b 2 c 3"

print(re.split(r"\d+", text))

text = "a 11 b 2 3 c 4"

print(re.split(r"\d+", text))
```

```
['a ', ' b ', ' c ', '']
['a ', ' b ', ' ', ' c ', '']
```

Text pre-processing steps

Text preprocessing involves transforming text into a clean and consistent format that can then be fed into a model for further analysis and learning. Raw text data might contain unwanted or unimportant text due to which our results might not give efficient accuracy, and might make it hard to understand and analyze.

The various text preprocessing steps are:

1. Tokenization.
2. Lower casing.
3. Stop words removal.

4. Stemming.
5. Lemmatization.

Before implementing the pre-processing, let us understand the concept first.

Tokenization

Tokenization is used in NLP to split paragraphs and sentences into smaller units that can be more easily assigned meaning.

The first step of the NLP process is gathering the data (a sentence) and breaking it into understandable parts (words).

- **Sentence Tokenization**

In [22]:
```
nltk.download('punkt_tab')
nltk.download('punkt') #Punkt is a pretrained unsupervised machine learning tokeniz
```

```
[nltk_data] Downloading package punkt_tab to
[nltk_data]     C:\Users\user\AppData\Roaming\nltk_data...
[nltk_data]   Unzipping tokenizers\punkt_tab.zip.
[nltk_data] Downloading package punkt to
[nltk_data]     C:\Users\user\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
```

Out[22]: True

In [23]:
```
import nltk

text = "I'm enjoying the NLP course! I am also learning new concepts."

print(nltk.sent_tokenize(text))
```

```
["I'm enjoying the NLP course!", 'I am also learning new concepts.']
```

- **Word Tokenzitaion**

In [24]:
```
text = "I'm enjoying the NLP course!"
text.split()
```

Out[24]: ["I'm", 'enjoying', 'the', 'NLP', 'course!']

In [25]:
```
import nltk

text = "I'm enjoying the NLP course!"

print(nltk.word_tokenize(text)) #I'm is tokenized to two tokens [I,m] course! is al
```

```
['I', "'m", 'enjoying', 'the', 'NLP', 'course', '!']
```

In [26]:
```
!pip install -U spacy
!python -m spacy download en_core_web_sm
```

```
Collecting spacy
  Downloading spacy-3.8.16-cp313-cp313-win_amd64.whl.metadata (28 kB)
Collecting spacy-legacy<3.1.0,>=3.0.11 (from spacy)
  Downloading spacy_legacy-3.0.12-py2.py3-none-any.whl.metadata (2.8 kB)
Collecting spacy-loggers<2.0.0,>=1.0.0 (from spacy)
  Downloading spacy_loggers-1.0.5-py3-none-any.whl.metadata (23 kB)
Collecting murmurhash<1.1.0,>=0.28.0 (from spacy)
  Downloading murmurhash-1.0.15-cp313-cp313-win_amd64.whl.metadata (2.3 kB)
Collecting cymem<2.1.0,>=2.0.2 (from spacy)
  Downloading cymem-2.0.13-cp313-cp313-win_amd64.whl.metadata (9.9 kB)
Collecting preshed<3.1.0,>=3.0.2 (from spacy)
  Downloading preshed-3.0.13-cp313-cp313-win_amd64.whl.metadata (5.4 kB)
Collecting thinc<8.4.0,>=8.3.12 (from spacy)
  Downloading thinc-8.3.13-cp313-cp313-win_amd64.whl.metadata (15 kB)
Collecting wasabi<1.2.0,>=0.9.1 (from spacy)
  Downloading wasabi-1.1.3-py3-none-any.whl.metadata (28 kB)
Collecting srsly<3.0.0,>=2.5.3 (from spacy)
  Downloading srsly-2.5.3-cp313-cp313-win_amd64.whl.metadata (20 kB)
Collecting catalogue<2.1.0,>=2.0.6 (from spacy)
  Downloading catalogue-2.0.10-py3-none-any.whl.metadata (14 kB)
Collecting weasel<2.0.0,>=1.0.0 (from spacy)
  Downloading weasel-1.0.0-py3-none-any.whl.metadata (4.6 kB)
Collecting confection<2.0.0,>=1.3.2 (from spacy)
  Downloading confection-1.3.3-py3-none-any.whl.metadata (19 kB)
Collecting typer<1.0.0,>=0.3.0 (from spacy)
  Downloading typer-0.27.2-py3-none-any.whl.metadata (16 kB)
Requirement already satisfied: click<9.0.0,>=8.2.1 in C:\Users\user\AppData\Local\Pr
ograms\Python\Python313\Lib\site-packages (from spacy) (8.5.0)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in C:\Users\user\AppData\Local\Pr
ograms\Python\Python313\Lib\site-packages (from spacy) (4.70.1)
Requirement already satisfied: numpy>=1.19.0 in C:\Users\user\AppData\Local\Programs
\Python\Python313\Lib\site-packages (from spacy) (2.4.4)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in C:\Users\user\AppData\Loca
l\Programs\Python\Python313\Lib\site-packages (from spacy) (2.34.2)
Collecting pydantic<3.0.0,>=2.0.0 (from spacy)
  Downloading pydantic-2.13.5-py3-none-any.whl.metadata (110 kB)
Requirement already satisfied: jinja2 in C:\Users\user\AppData\Local\Programs\Python
\Python313\Lib\site-packages (from spacy) (3.1.6)
Collecting setuptools (from spacy)
  Using cached setuptools-84.0.0-py3-none-any.whl.metadata (6.6 kB)
Requirement already satisfied: packaging>=20.0 in C:\Users\user\AppData\Roaming\Pyth
on\Python313\site-packages (from spacy) (26.0)
Collecting annotated-types>=0.6.0 (from pydantic<3.0.0,>=2.0.0->spacy)
  Downloading annotated_types-0.8.0-py3-none-any.whl.metadata (15 kB)
Collecting pydantic-core==2.46.5 (from pydantic<3.0.0,>=2.0.0->spacy)
  Downloading pydantic_core-2.46.5-cp313-cp313-win_amd64.whl.metadata (6.7 kB)
Requirement already satisfied: typing-extensions>=4.14.1 in C:\Users\user\AppData\Lo
cal\Programs\Python\Python313\Lib\site-packages (from pydantic<3.0.0,>=2.0.0->spacy)
(4.16.0)
Collecting typing-inspection>=0.4.2 (from pydantic<3.0.0,>=2.0.0->spacy)
  Downloading typing_inspection-0.4.4-py3-none-any.whl.metadata (2.6 kB)
Requirement already satisfied: charset_normalizer<4,>=2 in C:\Users\user\AppData\Loc
al\Programs\Python\Python313\Lib\site-packages (from requests<3.0.0,>=2.13.0->spacy)
(3.5.1)
Requirement already satisfied: idna<4,>=2.5 in C:\Users\user\AppData\Local\Programs
\Python\Python313\Lib\site-packages (from requests<3.0.0,>=2.13.0->spacy) (3.19)
```

```
Requirement already satisfied: urllib3<3,>=1.26 in C:\Users\user\AppData\Local\Progr
ams\Python\Python313\Lib\site-packages (from requests<3.0.0,>=2.13.0->spacy) (2.7.0)
Requirement already satisfied: certifi>=2023.5.7 in C:\Users\user\AppData\Local\Prog
rams\Python\Python313\Lib\site-packages (from requests<3.0.0,>=2.13.0->spacy) (202
6.7.22)
Collecting blis<1.4.0,>=1.3.0 (from thinc<8.4.0,>=8.3.12->spacy)
  Downloading blis-1.3.3-cp313-cp313-win_amd64.whl.metadata (7.7 kB)
Requirement already satisfied: colorama in C:\Users\user\AppData\Roaming\Python\Pyth
on313\site-packages (from tqdm<5.0.0,>=4.38.0->spacy) (0.4.6)
Collecting shellingham>=1.3.0 (from typer<1.0.0,>=0.3.0->spacy)
  Downloading shellingham-1.5.4-py2.py3-none-any.whl.metadata (3.5 kB)
Collecting rich>=13.8.0 (from typer<1.0.0,>=0.3.0->spacy)
  Downloading rich-15.0.0-py3-none-any.whl.metadata (18 kB)
Collecting annotated-doc>=0.0.2 (from typer<1.0.0,>=0.3.0->spacy)
  Downloading annotated_doc-0.0.5-py3-none-any.whl.metadata (6.5 kB)
Collecting cloudpathlib>=0.7.0 (from weasel<2.0.0,>=1.0.0->spacy)
  Downloading cloudpathlib-0.25.0-py3-none-any.whl.metadata (16 kB)
Collecting smart-open>=5.2.1 (from weasel<2.0.0,>=1.0.0->spacy)
  Downloading smart_open-8.0.1-py3-none-any.whl.metadata (24 kB)
Collecting httpx>=0.24.0 (from weasel<2.0.0,>=1.0.0->spacy)
  Downloading httpx-0.28.1-py3-none-any.whl.metadata (7.1 kB)
Collecting anyio (from httpx>=0.24.0->weasel<2.0.0,>=1.0.0->spacy)
  Downloading anyio-4.15.1-py3-none-any.whl.metadata (4.7 kB)
Collecting httpcore==1.* (from httpx>=0.24.0->weasel<2.0.0,>=1.0.0->spacy)
  Downloading httpcore-1.0.9-py3-none-any.whl.metadata (21 kB)
Collecting h11>=0.16 (from httpcore==1.*->httpx>=0.24.0->weasel<2.0.0,>=1.0.0->spac
y)
  Downloading h11-0.16.0-py3-none-any.whl.metadata (8.3 kB)
Collecting markdown-it-py>=2.2.0 (from rich>=13.8.0->typer<1.0.0,>=0.3.0->spacy)
  Downloading markdown_it_py-4.2.0-py3-none-any.whl.metadata (7.4 kB)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in C:\Users\user\AppData\Roam
ing\Python\Python313\site-packages (from rich>=13.8.0->typer<1.0.0,>=0.3.0->spacy)
(2.20.0)
Collecting mdurl~=0.1 (from markdown-it-py>=2.2.0->rich>=13.8.0->typer<1.0.0,>=
0.3.0->spacy)
  Downloading mdurl-0.1.2-py3-none-any.whl.metadata (1.6 kB)
Collecting wrapt (from smart-open>=5.2.1->weasel<2.0.0,>=1.0.0->spacy)
  Downloading wrapt-2.4.1-cp313-cp313-win_amd64.whl.metadata (7.6 kB)
Requirement already satisfied: MarkupSafe>=2.0 in C:\Users\user\AppData\Local\Progra
ms\Python\Python313\Lib\site-packages (from jinja2->spacy) (3.0.3)
Downloading spacy-3.8.16-cp313-cp313-win_amd64.whl (15.2 MB)
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    --------------------------------------- 0.3/15.2 MB ? eta -:--:--
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   ----------- ---------------------------- 4.2/15.2 MB 9.7 MB/s eta 0:00:02
   ------------------------------ ---------- 11.3/15.2 MB 16.6 MB/s eta 0:00:01
   ---------------------------------------- 15.2/15.2 MB 17.8 MB/s  0:00:01
Downloading catalogue-2.0.10-py3-none-any.whl (17 kB)
Downloading confection-1.3.3-py3-none-any.whl (35 kB)
Downloading cymem-2.0.13-cp313-cp313-win_amd64.whl (40 kB)
Downloading murmurhash-1.0.15-cp313-cp313-win_amd64.whl (25 kB)
Downloading preshed-3.0.13-cp313-cp313-win_amd64.whl (122 kB)
Downloading pydantic-2.13.5-py3-none-any.whl (472 kB)
Downloading pydantic_core-2.46.5-cp313-cp313-win_amd64.whl (2.0 MB)
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   ---------------------------------------- 2.0/2.0 MB 29.6 MB/s  0:00:00
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Downloading spacy_legacy-3.0.12-py2.py3-none-any.whl (29 kB)
Downloading spacy_loggers-1.0.5-py3-none-any.whl (22 kB)
Downloading srsly-2.5.3-cp313-cp313-win_amd64.whl (650 kB)
   ---------------------------------------- 0.0/650.8 kB ? eta -:--:--
   ---------------------------------------- 650.8/650.8 kB 17.3 MB/s  0:00:00
Downloading thinc-8.3.13-cp313-cp313-win_amd64.whl (1.7 MB)
   ---------------------------------------- 0.0/1.7 MB ? eta -:--:--
   ---------------------------------------- 1.7/1.7 MB 20.6 MB/s  0:00:00
Downloading blis-1.3.3-cp313-cp313-win_amd64.whl (6.2 MB)
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   ---------------------------------------- 6.2/6.2 MB 27.3 MB/s  0:00:00
Downloading typer-0.27.2-py3-none-any.whl (123 kB)
Downloading wasabi-1.1.3-py3-none-any.whl (27 kB)
Downloading weasel-1.0.0-py3-none-any.whl (50 kB)
Downloading annotated_doc-0.0.5-py3-none-any.whl (5.3 kB)
Downloading annotated_types-0.8.0-py3-none-any.whl (13 kB)
Downloading cloudpathlib-0.25.0-py3-none-any.whl (66 kB)
Downloading httpx-0.28.1-py3-none-any.whl (73 kB)
Downloading httpcore-1.0.9-py3-none-any.whl (78 kB)
Downloading h11-0.16.0-py3-none-any.whl (37 kB)
Downloading rich-15.0.0-py3-none-any.whl (310 kB)
Downloading markdown_it_py-4.2.0-py3-none-any.whl (91 kB)
Downloading mdurl-0.1.2-py3-none-any.whl (10.0 kB)
Downloading shellingham-1.5.4-py2.py3-none-any.whl (9.8 kB)
Downloading smart_open-8.0.1-py3-none-any.whl (73 kB)
Downloading typing_inspection-0.4.4-py3-none-any.whl (14 kB)
Downloading anyio-4.15.1-py3-none-any.whl (132 kB)
Using cached setuptools-84.0.0-py3-none-any.whl (818 kB)
Downloading wrapt-2.4.1-cp313-cp313-win_amd64.whl (98 kB)
Installing collected packages: wrapt, wasabi, typing-inspection, spacy-loggers, spac
y-legacy, shellingham, setuptools, pydantic-core, murmurhash, mdurl, h11, cymem, con
fection, cloudpathlib, catalogue, blis, anyio, annotated-types, annotated-doc, srsly
, smart-open, pydantic, preshed, markdown-it-py, httpcore, thinc, rich, httpx, typer
, weasel, spacy

   ----------------------------------------  0/31 [wrapt]
   ----------------------------------------  0/31 [wrapt]
   ----------------------------------------  0/31 [wrapt]
   - --------------------------------------  1/31 [wasabi]
   - --------------------------------------  1/31 [wasabi]
   - --------------------------------------  1/31 [wasabi]
   - --------------------------------------  1/31 [wasabi]
   -- -------------------------------------  2/31 [typing-inspection]
   -- -------------------------------------  2/31 [typing-inspection]
   --- ------------------------------------  3/31 [spacy-loggers]
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  WARNING: Retrying (Retry(total=4, connect=None, read=None, redirect=None, status=N
one)) after connection broken by 'ProtocolError('Connection aborted.', ConnectionRes
etError(10054, 'An existing connection was forcibly closed by the remote host', None
, 10054, None))': /explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_
core_web_sm-3.8.0-py3-none-any.whl
  WARNING: Retrying (Retry(total=3, connect=None, read=None, redirect=None, status=N
one)) after connection broken by 'ProtocolError('Connection aborted.', ConnectionRes
etError(10054, 'An existing connection was forcibly closed by the remote host', None
, 10054, None))': /explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_
core_web_sm-3.8.0-py3-none-any.whl
  WARNING: Retrying (Retry(total=2, connect=None, read=None, redirect=None, status=N
one)) after connection broken by 'ProtocolError('Connection aborted.', ConnectionRes
etError(10054, 'An existing connection was forcibly closed by the remote host', None
, 10054, None))': /explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_
core_web_sm-3.8.0-py3-none-any.whl
  WARNING: Retrying (Retry(total=1, connect=None, read=None, redirect=None, status=N
one)) after connection broken by 'ProtocolError('Connection aborted.', ConnectionRes
etError(10054, 'An existing connection was forcibly closed by the remote host', None
, 10054, None))': /explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_
core_web_sm-3.8.0-py3-none-any.whl
  WARNING: Retrying (Retry(total=0, connect=None, read=None, redirect=None, status=N
one)) after connection broken by 'ProtocolError('Connection aborted.', ConnectionRes
etError(10054, 'An existing connection was forcibly closed by the remote host', None
, 10054, None))': /explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_
core_web_sm-3.8.0-py3-none-any.whl
ERROR: Could not install packages due to an OSError: HTTPSConnectionPool(host='githu
b.com', port=443): Max retries exceeded with url: /explosion/spacy-models/releases/d
ownload/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any.whl (Caused by Protoc
olError('Connection aborted.', ConnectionResetError(10054, 'An existing connection w
as forcibly closed by the remote host', None, 10054, None)))


[notice] A new release of pip is available: 26.0.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

In [31]:
```
!python -m spacy download en_core_web_sm
```

```
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_we
b_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any.whl (12.8 MB)
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     ---------------------------------------- 12.8/12.8 MB 53.8 kB/s  0:05:26
Installing collected packages: en-core-web-sm
Successfully installed en-core-web-sm-3.8.0
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
```

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  WARNING: Retrying (Retry(total=4, connect=None, read=None, redirect=None, status=N
one)) after connection broken by 'NewConnectionError('<pip._vendor.urllib3.connectio
n.HTTPSConnection object at 0x0000013DF570DBE0>: Failed to establish a new connectio
n: [Errno 11001] getaddrinfo failed')': /explosion/spacy-models/releases/download/en
_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any.whl
  WARNING: Retrying (Retry(total=3, connect=None, read=None, redirect=None, status=N
one)) after connection broken by 'ReadTimeoutError("HTTPSConnectionPool(host='githu
b.com', port=443): Read timed out. (read timeout=15)")': /explosion/spacy-models/rel
eases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any.whl
  WARNING: Retrying (Retry(total=4, connect=None, read=None, redirect=None, status=N
one)) after connection broken by 'NewConnectionError('<pip._vendor.urllib3.connectio
n.HTTPSConnection object at 0x0000013DF56EB110>: Failed to establish a new connectio
n: [Errno 11001] getaddrinfo failed')': /github-production-release-asset/84940268/15
132ab6-4050-4914-8fe8-ac2c2fdcb9cf?sp=r&sv=2018-11-09&sr=b&spr=https&se=2026-09-15T0
7%3A54%3A46Z&rscd=attachment%3B+filename%3Den_core_web_sm-3.8.0-py3-none-any.whl&rsc
t=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a66
54-997b-47e9-b12b-9515b896b4de&skt=2026-09-15T06%3A54%3A29Z&ske=2026-09-15T07%3A54%3
A46Z&sks=b&skv=2018-11-09&sig=5tI40d4Q1vKbbteklIU3psXpZz3PgocY3crphgEL%2B5c%3D&jwt=e
yJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiYXVkIjoicmVsZWFzZS1h
c3NldHMuZ2l0aHVidXNlcmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc4OTQ1NzgxOSwibmJm
IjoxNzg5NDU2MDE5LCJwYXRoIjoicmVsZWFzZWFzZ2V0cHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy5u
ZXQifQ.-lVJ2K5oSsAjXkqWwStcI1XTZd-eQhp9WYNGBOt0Qug&response-content-disposition=atta
chment%3B%20filename%3Den_core_web_sm-3.8.0-py3-none-any.whl&response-content-type=a
pplication%2Foctet-stream
  WARNING: Retrying (Retry(total=3, connect=None, read=None, redirect=None, status=N
one)) after connection broken by 'NewConnectionError('<pip._vendor.urllib3.connectio
n.HTTPSConnection object at 0x0000013DF56EB390>: Failed to establish a new connectio
n: [Errno 11001] getaddrinfo failed')': /github-production-release-asset/84940268/15
132ab6-4050-4914-8fe8-ac2c2fdcb9cf?sp=r&sv=2018-11-09&sr=b&spr=https&se=2026-09-15T0
7%3A54%3A46Z&rscd=attachment%3B+filename%3Den_core_web_sm-3.8.0-py3-none-any.whl&rsc
t=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a66
54-997b-47e9-b12b-9515b896b4de&skt=2026-09-15T06%3A54%3A29Z&ske=2026-09-15T07%3A54%3
A46Z&sks=b&skv=2018-11-09&sig=5tI40d4Q1vKbbteklIU3psXpZz3PgocY3crphgEL%2B5c%3D&jwt=e
yJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiYXVkIjoicmVsZWFzZS1h
c3NldHMuZ2l0aHVidXNlcmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc4OTQ1NzgxOSwibmJm
IjoxNzg5NDU2MDE5LCJwYXRoIjoicmVsZWFzZWFzZ2V0cHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy5u
ZXQifQ.-lVJ2K5oSsAjXkqWwStcI1XTZd-eQhp9WYNGBOt0Qug&response-content-disposition=atta
chment%3B%20filename%3Den_core_web_sm-3.8.0-py3-none-any.whl&response-content-type=a
pplication%2Foctet-stream
  WARNING: Retrying (Retry(total=2, connect=None, read=None, redirect=None, status=N
one)) after connection broken by 'NewConnectionError('<pip._vendor.urllib3.connectio
n.HTTPSConnection object at 0x0000013DF56EB610>: Failed to establish a new connectio
n: [Errno 11001] getaddrinfo failed')': /github-production-release-asset/84940268/15
132ab6-4050-4914-8fe8-ac2c2fdcb9cf?sp=r&sv=2018-11-09&sr=b&spr=https&se=2026-09-15T0
7%3A54%3A46Z&rscd=attachment%3B+filename%3Den_core_web_sm-3.8.0-py3-none-any.whl&rsc
t=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a66
54-997b-47e9-b12b-9515b896b4de&skt=2026-09-15T06%3A54%3A29Z&ske=2026-09-15T07%3A54%3
A46Z&sks=b&skv=2018-11-09&sig=5tI40d4Q1vKbbteklIU3psXpZz3PgocY3crphgEL%2B5c%3D&jwt=e
yJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiYXVkIjoicmVsZWFzZS1h
c3NldHMuZ2l0aHVidXNlcmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc4OTQ1NzgxOSwibmJm
IjoxNzg5NDU2MDE5LCJwYXRoIjoicmVsZWFzZWFzZ2V0cHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy5u
ZXQifQ.-lVJ2K5oSsAjXkqWwStcI1XTZd-eQhp9WYNGBOt0Qug&response-content-disposition=atta
chment%3B%20filename%3Den_core_web_sm-3.8.0-py3-none-any.whl&response-content-type=a
pplication%2Foctet-stream

[notice] A new release of pip is available: 26.0.1 -> 26.2.1
```

```
[notice] To update, run: python.exe -m pip install --upgrade pip
```

In [32]:
```
import spacy
nlp = spacy.load('en_core_web_sm') # Load the small English model - - nlp becomes a

doc = nlp("I'm enjoying the NLP course!") # Process the text using nlp model, retur

for token in doc: # Iterate over each token in the Doc object.
    print(token)
```

```
I
'm
enjoying
the
NLP
course
!
```

In [33]:
```
text = "A 5km NYC cab ride costs $10.30"
doc = nlp("A 5km NYC cab ride costs $10.30")

print('nltk Tokenizer result: ')
print(nltk.word_tokenize(text))

print('\nSpaCy Tokenizer result: ')

for token in doc:
    print(token)
```

```
nltk Tokenizer result:
['A', '5km', 'NYC', 'cab', 'ride', 'costs', '$', '10.30']

SpaCy Tokenizer result:
A
5
km
NYC
cab
ride
costs
$
10.30
```

Lower Casing

Converting a word to lower case (NLP -> nlp). Words like Book and book mean the same but when not converted to the lower case those two are represented as two different words in the vector space model (resulting in more dimensions).

In [34]:
```
text = "I'm enjoying the NLP course!"
text = text.lower()
text
```

Out[34]:
```
"i'm enjoying the nlp course!"
```

Stemming

Stemming is basically removing the suffix from a word and reduce it to its root word. For example: “Flying” is a word and its suffix is “ing”, if we remove “ing” from “Flying” then we will get base word or root word which is “Fly”. We uses these suffix to create a new word from original stem word.

https://www.nltk.org/howto/stem.html

What is PorterStemmer?

It is one of the most popular stemming methods proposed in 1980. It is based on the idea that the suffixes in the English language are made up of a combination of smaller and simpler suffixes. This stemmer is known for its speed and simplicity. The main applications of Porter Stemmer include data mining and Information retrieval. However, its applications are only limited to English words. Also, the group of stems is mapped on to the same stem and the output stem is not necessarily a meaningful word. The algorithms are fairly lengthy in nature and are known to be the oldest stemmer.

> Advantage: It produces the best output as compared to other stemmers and it has less error rate.

> Limitation: Morphological variants produced are not always real words.

In [35]:
```
import nltk
from nltk.stem import PorterStemmer

ps = PorterStemmer() # create a PorterStemmer object

words = ['run','runner','running','ran','runs','easily','fairly']

for word in words:
    print(word + ' --> ' + ps.stem(word))
```

```
run --> run
runner --> runner
running --> run
ran --> ran
runs --> run
easily --> easili
fairly --> fairli
```

What is SnowballStemmer?

It is a stemming algorithm which is also known as the Porter2 stemming algorithm as it is a better version of the Porter Stemmer since some issues of it were fixed in this stemmer.

> Advantage: It is slightly faster computation time than porter, with a reasonably large community around it.

In [36]:
```
import nltk
from nltk.stem.snowball import SnowballStemmer

sn = SnowballStemmer(language='english')

words = ['run','runner','running','ran','runs','easily','fairly']

for word in words:
    print(word+' --> '+sn.stem(word))
```

```
run --> run
runner --> runner
running --> run
ran --> ran
runs --> run
easily --> easili
fairly --> fair
```

Comparing between porter and snowball

In [37]:
```
words = ['generous','generation','generously','generate']

for word in words:
    print(word+' --> '+ps.stem(word))
    print(word+' --> '+sn.stem(word))
    print('----------------------------------------')
```

```
generous --> gener
generous --> generous
----------------------------------------
generation --> gener
generation --> generat
----------------------------------------
generously --> gener
generously --> generous
----------------------------------------
generate --> gener
generate --> generat
----------------------------------------
```

Note: Spacy does not provide stemming

Lemmatization

Lemmatization is the process of grouping together the different inflected forms of a word so

they can be analyzed as a single item. Lemmatization is similar to stemming but it brings context to the words. So it links words with similar meanings to one word.

> One major difference with stemming is that lemmatize takes a part of speech parameter (pos) If not supplied, the default is "noun."

Part of Speech	Description	Example Words
Noun (N)	Person, place, thing, idea	dog, city, love, Python
Verb (V)	Action or state	run, eat, is, was, jump
Adjective (ADJ)	Describes a noun	happy, blue, fast
Adverb (ADV)	Describes a verb/adjective	quickly, very, well
Pronoun (PRON)	Replaces a noun	he, she, it, they
Preposition (ADP)	Shows relationship	in, on, under, with
Conjunction (CONJ)	Connects words/clauses	and, but, or
Determiner (DET)	Introduces a noun	a, an, the
Interjection (INTJ)	Expresses emotion	wow, oh, hey

In [38]:
```
nltk.download('wordnet')
```

```
[nltk_data] Downloading package wordnet to
[nltk_data]     C:\Users\user\AppData\Roaming\nltk_data...
```

Out[38]: True

In [39]:
```
import nltk
from nltk.stem import WordNetLemmatizer

text = 'I am enjoying AI courses. I am taking NLP course.'
lemmatizer = WordNetLemmatizer()

for word in nltk.word_tokenize(text):
    print(f"{word}: ", lemmatizer.lemmatize(word))
```

```
I:  I
am:  am
enjoying:  enjoying
AI:  AI
courses:  course
.:  .
I:  I
am:  am
taking:  taking
NLP:  NLP
course:  course
.:  .
```

In [40]:
```
# Changing pos to verb
```

```
for word in nltk.word_tokenize(text):
    print(f"{word}: ", lemmatizer.lemmatize(word, pos='v'))
```

```
I:  I
am:  be
enjoying:  enjoy
AI:  AI
courses:  course
.:  .
I:  I
am:  be
taking:  take
NLP:  NLP
course:  course
.:  .
```

You may go through Spacy's lemmatizer: https://spacy.io/api/lemmatizer

Stop Words Removal

The words which are generally filtered out before processing a natural language are called stop words. These are actually the most common words in any language (like articles, prepositions, pronouns, conjunctions, etc) and does not add much information to the text. Examples of a few stop words in English are “the”, “a”, “an”, “so”, “what”.

Why do we need to remove stopwords?

By removing these words, we remove the low-level information from our text in order to give more focus to the important information. In order words, we can say that the removal of such words does not show any negative consequences on the model we train for our task.

- reduces the dataset size
- reduces the training time

Do we always remove stopwords? **NO!**

We do not always remove the stop words. The removal of stop words is highly dependent on the task we are performing and the goal we want to achieve. For example, if we are training a model that can perform the sentiment analysis task, we might not remove the stop words.

Movie review: “The movie was not good at all.”

Text after removal of stop words: “movie good”

In [41]:
```
import spacy
nlp = spacy.load('en_core_web_sm')
```

In [42]:
```
print(nlp.Defaults.stop_words)
print(len(nlp.Defaults.stop_words))
```

```
{'amount', 'this', 'via', 'a', 'whence', 'other', 'sixty', 'our', 'therefore', 'coul
d', 'few', 'some', 'am', 'therein', 'fifteen', 'somewhere', 'ten', 'put', 'several',
'afterwards', 'everywhere', 'n’t', 'namely', 'none', 'or', 'yourselves', '‘m', 'besi
des', 'really', "'d", 'can', 'alone', 'well', 'also', 'someone', 'among', 'had',
'‘d', 'make', 'he', 're', 'those', 'above', 'mine', 'beside', 'and', 'us', 'through'
, 'eight', 'anyone', 'hers', 'whenever', 'see', 'never', 'where', 'did', '’m', 'move
', 'they', 'toward', "'m", 'its', 'used', 'own', 'against', 'themselves', 'whether',
'hereafter', 'become', 'show', 'be', 'name', 'fifty', 'have', 'towards', 'often', 'h
ere', 'when', 'off', 'over', 'all', 'itself', 'sometime', 'an', 'would', 'although',
'hereby', 'two', 'these', 'down', 'four', 'within', 'does', 'bottom', 'until', 'his'
, 'nor', 'ours', 'it', 'seemed', 'is', 'mostly', 'many', 'i', 'cannot', 'whatever',
'your', 'forty', 'both', 'say', 'while', 'rather', 'doing', 'just', 'moreover', 'bec
ause', 'call', 'whereafter', 'hence', 'third', 'thence', 'herein', '’re', 'what', 'd
o', 'most', 'wherever', 'sometimes', 'there', 'seems', 'hereupon', '’s', 'around', '
onto', 'at', 'side', 'ever', 'in', '‘s', 'though', 'ca', 'thru', 'nevertheless', 'an
ything', 'too', 'anywhere', 'empty', 'as', 'seeming', 'been', 'done', 'across', 'but
', '‘ll', 'hundred', "n't", 'which', 'more', 'of', 'me', 'whereby', 'first', 'each',
'quite', 'must', 'up', 'amongst', 'only', 'below', 'front', 'keep', 'since', 'she',
'nowhere', 'thereupon', 'whose', 'will', 'by', 'nobody', 'together', 'everyone', 'we
re', 'again', 'whoever', 'whom', 'give', 'last', 'upon', 'between', 'enough', 'might
', 'before', 'nine', 'with', 'to', 'whereas', 'out', '’ll', "'ve", 'nothing', 'where
in', 'such', 'behind', 'somehow', 'along', 'whither', "'s", 'still', 'former', "'ll"
, 'top', 'made', 'her', 'five', 'full', 'if', 'yet', 'for', 'their', 'became', 'go',
'elsewhere', 'even', 'however', 'ourselves', 'something', 'you', 'twelve', 'please',
'anyway', 'once', 'noone', 'serious', 'latter', 'per', 'no', 'throughout', 'regardin
g', 'one', 'unless', 'are', 'beyond', 'how', 'about', 'back', 'whereupon', 'perhaps'
, 'neither', 'same', 'without', 'using', 'why', 'yours', 'either', 'becoming', 'exce
pt', '’d', 'should', '’ve', 'any', 'due', 'less', 'almost', 'was', 'next', 'them', '
into', 'six', 'himself', 'not', 'indeed', 'now', 'then', 'least', 'him', 'seem', 'th
us', 'so', 'thereby', 'twenty', 'formerly', 'thereafter', 'part', 'another', 'every'
, 'may', 'herself', 'under', 'than', '‘ve', 'very', 'during', 'who', 'whole', '‘re',
'take', 'my', 'we', 'eleven', 'the', 'has', 'myself', 'that', 'beforehand', 'much',
'various', 'n‘t', 'three', 'being', "'re", 'always', 'everything', 'anyhow', 'latter
ly', 'else', 'meanwhile', 'otherwise', 'becomes', 'yourself', 'on', 'get', 'further'
, 'already', 'after', 'from', 'others'}
326
```

In [43]:
```
print(nlp.vocab['myself'].is_stop)
print(nlp.vocab['mystery'].is_stop)
```

```
True
False
```

In [44]:
```
print (nlp.vocab['btw'].is_stop) #print False
```

```
False
```

In [45]:
```
nlp.Defaults.stop_words.add('btw')

nlp.vocab['btw'].is_stop = True # You must update the is_stop flag for the word you

print (nlp.vocab['btw'].is_stop)
```

```
True
```

In [46]:
```
nlp.Defaults.stop_words.remove('beyond')
nlp.vocab['beyond'].is_stop = False
```

```
print(nlp.vocab['beyond'].is_stop)
```

```
False
```

Stop words in NLTK

In [47]:
```
import nltk
nltk.download('stopwords')
```

```
[nltk_data] Downloading package stopwords to
[nltk_data]     C:\Users\user\AppData\Roaming\nltk_data...
[nltk_data]   Unzipping corpora\stopwords.zip.
```

Out[47]: True

In [48]:
```
from nltk.corpus import stopwords

stop_words = set(stopwords.words('english'))
print(len(stop_words))
```

```
198
```

Text pre-processing Example

To make you understand better about the need of text pre-processing. We will do a small project. In this project, I need to find how many hashtags and the top 10 hashtags used in the dataset.

The dataset is about people tweets about apple company in the twitter. You can find the dataset here:

https://www.kaggle.com/datasets/seriousran/appletwittersentimenttexts

In [ ]:
```
import kagglehub


path = kagglehub.dataset_download("seriousran/appletwittersentimenttexts")

print("Path to dataset files:", path)
```

```
c:\Users\user\AppData\Local\Programs\Python\Python313\Lib\site-packages\tqdm\auto.p
y:21: TqdmWarning: IProgress not found. Please update jupyter and ipywidgets. See ht
tps://ipywidgets.readthedocs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm
Downloading to C:\Users\user\.cache\kagglehub\datasets\seriousran\appletwittersentim
enttexts\1.archive...
100%|██████████| 71.0k/71.0k [00:00<00:00, 5.31MB/s]
Extracting files...
Path to dataset files: C:\Users\user\.cache\kagglehub\datasets\seriousran\appletwitt
ersentimenttexts\versions\1
```

In [50]:
```
import pandas as pd
df = pd.read_csv("apple-twitter-sentiment-texts.csv")
```

In [ ]:
```
import re

hashtags = []

for tweet in df['text']:
    hashtags.extend(re.findall(r'#\w+', str(tweet).lower()))


print("Total hashtags:", len(hashtags))


hashtag_counts = {}

for hashtag in hashtags:
    if hashtag in hashtag_counts:
        hashtag_counts[hashtag] += 1
    else:
        hashtag_counts[hashtag] = 1


sorted_hashtags = sorted(
    hashtag_counts.items(),
    key=lambda x: x[1],
    reverse=True
)


print("\nTop 10 hashtags:")
for hashtag, count in sorted_hashtags[:10]:
    print(hashtag, count)
```

```
Total hashtags: 1616

Top 10 hashtags:
#aapl 400
#apple 162
#iphone 38
#december 37
#iphone6 29
#stocks 15
#ipad 14
#iphone6plus 13
#iphone6s 13
#tech 12
```